

RACHIT KHURANA

Voice AI & Full Stack Engineer building low-latency speech, desktop, mobile, and AI applications.

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SUMMARY

Engineer specializing in real-time speech-to-text systems, LLM-powered voice agents, and cross-platform desktop applications. Experienced building production AI products with Rust, Tauri, Python, React Native, and OpenAI/Deepgram APIs, with a focus on low-latency audio pipelines and scalable backend systems.

TECHNICAL SKILLS

Languages: Python, Rust, Golang, TypeScript, JavaScript, Kotlin

Voice AI / ML: OpenAI APIs, Deepgram, Whisper, Streaming Speech-to-Text (ASR), Text-to-Speech, Function Calling, Prompt Engineering, Vector Search / RAG

Frameworks: Tauri, React Native, React, Next.js, Django, FastAPI, Flask, Tailwind CSS

Backend & Databases: PostgreSQL, MongoDB, MySQL, Redis, WebSockets

Developer Tools: Docker, Git, Linux, Bash, Azure, GitHub Actions

EXPERIENCE

Hyathi Technologies

Full Stack Engineer

Jul 2026 – Present

Full Stack & Web3 Developer Intern

Dec 2024 – Jun 2026

- **Led development of BossAI's Windows desktop app and Android AI keyboard**, shipping a production voice-first typing experience across desktop and mobile.
- Built the Windows client using **Rust + Tauri**, delivering a **sub-30 MB installer** while maintaining runtime memory usage below **100 MB**.
- Built a native Android keyboard (**Kotlin IME**) that enables real-time voice typing, AI rewriting, grammar correction, and contextual text generation inside any application.
- Engineered a low-latency streaming speech pipeline with ASR, interim/final transcript merging, endpoint detection, punctuation restoration, contextual formatting, and instant text insertion.
- Built backend services in Node.js and Python for AI inference, authentication, transcription history, and production APIs supporting **600+ hours of real-world speech-to-text usage**.
- Implemented production features for a Canadian fintech platform using React Native, Node.js, MongoDB, and REST APIs, and developed retrieval pipelines for contextual AI responses.

PROJECTS

BossAI — Voice AI Keyboard & Desktop Assistant

Rust, Tauri, Python, React, Deepgram, OpenAI

- Built a cross-platform voice keyboard that converts natural speech into polished text inside any application.
- Implemented streaming transcription with interim results, endpoint detection, grammar correction, contextual formatting, and low-latency UI updates.
- Designed a desktop voice agent capable of executing natural language commands such as drafting emails, replying to messages, and generating structured content.

Ryla — AI Project Management Agent

Python, Next.js, Convex, Slack APIs, Basecamp APIs

- Built an AI agent that allows users to query projects, create tasks etc. from Basecamp directly within Slack conversations.
- Implemented conversational tool calling & retrieval across project tasks, discussions, and schedules.

DevCompete

Django, Redis, Docker, WebSockets

- Developed a real-time coding competition platform with isolated code execution & WebSocket-based live progress tracking.
- Built an AI-powered practice mode for coding challenge generation and interactive feedback.

ACHIEVEMENTS

- Winner — Coinbase Developer Track, EthIndia
- 3rd Place — Hedera Hello Future Hackathon (AI Track)
- Microsoft Azure Fundamentals (AZ-900)
- Winner — Onchain Summer Buildathon
- AWS Certified Cloud Practitioner

EDUCATION

Bennett University

B.Tech in Computer Science Engineering CGPA: 8.48/10

Greater Noida, India

2022 – 2026